SANTA CLARA UNIVERSITY • DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSEN 281: Pattern Recognition & Data Mining

Capstone Project Specification: US Wildfires

Duration: October 23 - November 30 Team: 2 students per project Total Points: 100

1. Overview

Welcome to your Capstone Project! You will transform USDA's geospatial wildfire event data into useful insights using the concepts covered in this course.

The project features a **flexible, open-ended framework** where your team will frame two core research questions: one addressed via a **Supervised** learning pipeline (Classification or Regression) and one via an **Unsupervised** learning pipeline (Clustering or Association Analysis).

Project Timeline & Milestones

1. **Capstone Milestone 1 (Due Fri 10/30):** Exploratory Data Analysis (EDA) & Problem Formulation.
2. **Capstone Milestone 2 (Due Fri 11/06):** Supervised Pipeline in place + preliminary modeling results.
3. **Capstone Milestone 3 (Due Fri 11/13):** Unsupervised Pipeline in place + preliminary pattern results.
4. **Optimization Phase (11/14 – 11/29):** Iterative feature engineering, algorithmic tuning, and hyperparameter optimization.
5. **Capstone Results Freeze (Mon 11/30):** Final model weights, results, and code frozen. No further training.
6. **Final Presentations & Demo (Wed 12/02):** In-class live code demonstration and presentation (slides optional).

Teams may consist of up to 2 students.

2. Database

The **Fire Program Analysis - Fire Occurrence Database (FPA-FOD)** is a scientific repository curated by research ecologist Dr. Karen Short and maintained by the **USDA Forest Service Research Data Archive**. The 2026 7th Edition (FPA_FOD_20260615.sqlite) bridges historical records from **1992 through 2024**, aggregating raw telemetry from federal, state, tribal, and local fire management organizations into a single, unified database engine. This is an engineered, rectangular database containing **2.66 million unique, geo-referenced wildfire incidents** that collectively represent **209 million acres burned** across a 33-year continuum. Every row in this database represents a distinct, declared wildfire incident.

Your **starter notebook** includes a baseline schema from the database. You are welcome to extract more columns from the SQLite engine as your project evolves:

1. **Fire Attributes**
   a. `'FIRE_NAME', 'COMPLEX_NAME'`

b. `'FIRE_SIZE'`
   Absolute final acreage consumed by a fire.
c. `'FIRE_SIZE_CLASS'`
   A categorical label (A to G) standardizing the scale of the disaster.
d. `'NWCG_CAUSE_CLASSIFICATION'`
   Human/Natural/Undetermined
e. `'NWCG_GENERAL_CAUSE'`
   Verified ignition sources characterizing human operations vs. natural elements (e.g., *Lightning, Fireworks, Arson, Equipment/Vehicle use, Power Lines*)

2. **Temporal attributes**
   a. `'FIRE_YEAR'`
   b. `'DISCOVERY_DATE', 'DISCOVERY_DOY', 'DISCOVERY_TIME',`
   c. `'CONT_DATE', 'CONT_DOY', 'CONT_TIME'`

   DISCOVERY_* is when the fire was discovered, and CONT_* is when it was contained.
   *_DATE = a format called **Julian Day Numbers -** continuous linear floats representing fractional days elapsed since a fixed epoch. Subtracting them yields a legitimate number of days [Julian Date Converter].
   *_DOY = Day-of-Year integers (1 to 365) - this format helps identify seasonal trends.
   *_TIME = 4-character hour-minute string markers HHMM.

3. **Geospatial Attributes**
   a. `'LATITUDE', 'LONGITUDE', 'STATE', 'COUNTY', 'FIPS_NAME', 'FIPS_CODE'`
      fips_name = name of county
   b. `'OWNER_DESCR'`
      Organization responsible for managing the land where the wildfire first ignited.

4. **Reporting Agency Attributes**
   a. `'NWCG_REPORTING_AGENCY'`
      Agency that filled out the fire incident report.
   b. `'SOURCE_SYSTEM_TYPE'`
      Source of fire record.

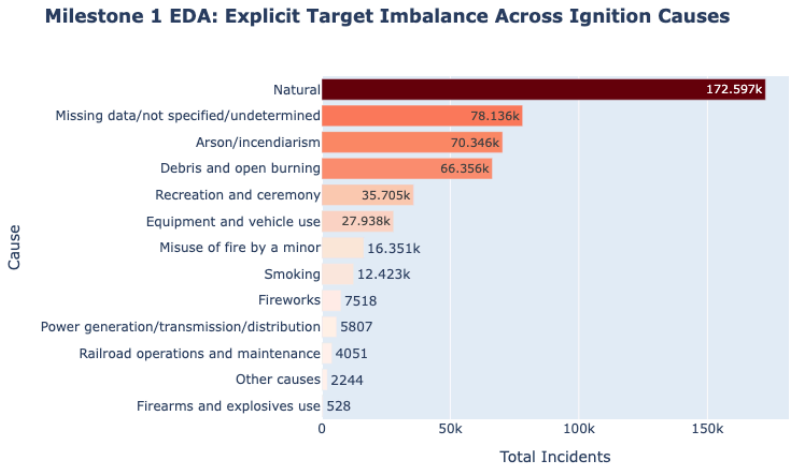
3. Frame the Data Mining Problems

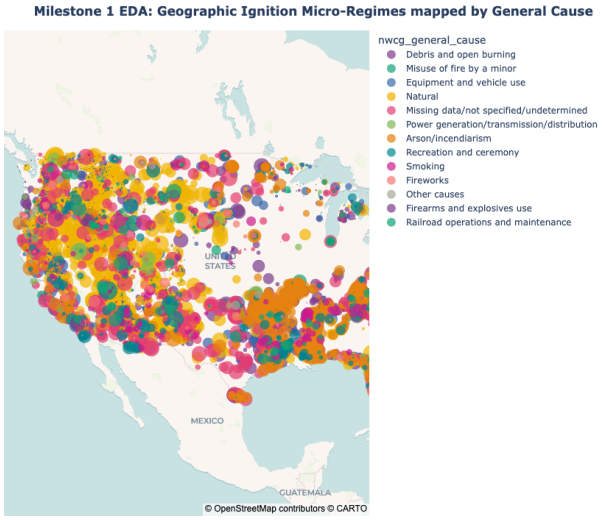
You will explore questions around wildfires that can be addressed by these Data Columns.

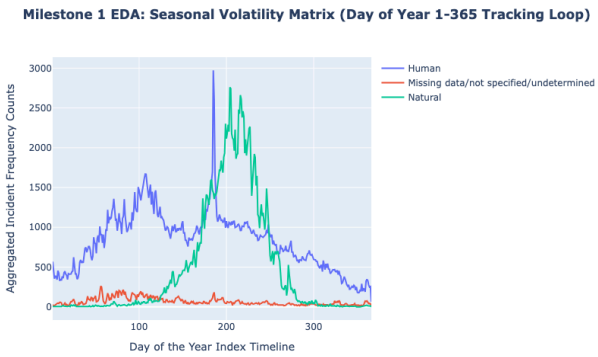
Examples of Data Mining Questions

- What is the primary cause of wildfires, and where do they usually occur?
- When do wildfires occur, and when are they usually discovered?
- Are there regional differences between wildfires across the US states?
- Do different agencies report fires differently?
- What does the fire size depend on?
- Or add your questions based on available data columns…

You will start with Exploratory Data Analysis to evaluate the viability of your questions - before doing any actual Data Mining. Example EDA plots are shown below. You should explore the data yourself towards answering questions of interest to you - frame one question as a Supervised Data Mining problem and one question as an Unsupervised Data Mining problem. Share the final EDA results and selected problems at Milestone 1.







4. Deliverables and Timeline

	Milestone 1 Due 10/30/26	Milestone 2 Due 11/6/26	Milestone 3 Due 11/13/26	Final Demo Due 12/2/26
Task	**Exploratory Data Analysis**	**Supervised Data Mining**	**Unsupervised Data Mining**	**Presentation and Demo in class**
Details	Visual analysis, problem definition	Classification, Regression	Clustering, Association	Live code demo, optional slides

			Analysis	
Deliverable	Functional .pynb with EDA plots + observations in markdown	Functional .pynb with performance plots + observations in markdown	Functional .pynb with performance plots + observations in markdown	Functional .pynb with performance plots + observations in markdown

5. Environment

Use your personal device if it works reasonably well - otherwise use HPC WAVE. The following instructions are not new - if you have already installed these, you can skip steps 1 and 2.

1. Follow Sarah's instructions to download miniconda
    a. Go to https://docs.conda.io/en/latest/miniconda.html and download the installer for your OS (Windows/Mac/Linux). Run it and accept the defaults.
2. Create class environment to download packages into (Python 3.13, Jupyterlab):
    a. `% conda env remove -n csen281`
    b. `% conda create -n csen281 -c conda-forge python=3.13 jupyterlab`
3. Run on your laptop
    a. `% conda activate csen281`
    b. `% conda install <libraries you need>`
    c. `% conda list`
    d. `% jupyter lab &`

4. Grading Rubric

- Tier 1: Exceptional (90% – 100% of Milestone Value)
    - Displays complete technical mastery, critical thinking, and publication-ready polish.
- Tier 2: Satisfactory (80% – 89% of Milestone Value)
    - Solid, baseline graduate-level work. The pipeline is functional, correct, and complete.
- Tier 3: Unsatisfactory (<80% of Milestone Value)
    - The deliverable contains fundamental design flaws, major gaps, or conceptual blind spots.

The points per Milestone and more are shown below. Your guidelines for each Milestone are also listed - they are not requirements and will NOT be used for rigorous grading.

Category	Guidelines	Points
Milestone 1 EDA & Framing	• Data Integrity & Artifact Profiling: Successful ingestion of the SQLite data engine. Identification and analytical visualization of reporting artifacts (e.g., coordinate rounding at county centers,	20

	institutional missingness in columns like DISCOVERY_TIME across SOURCE_SYSTEM_TYPE). ● Multi-Dimensional Profiling: Must show a minimum of three deep exploratory analysis sweeps: Temporal (e.g., extracting the "weekend effect" or seasonal diurnal paths), Spatial (e.g., hexbinning or spatial density maps avoiding raw point overplotting), and Administrative Scale (e.g., conditional boxplots or log-scale variations of FIRE_SIZE against OWNER_DESCR or NWCG_GENERAL_CAUSE). ● Rigorous Problem Formulation: Clear, unambiguous documentation of the selected data mining problems. The Supervised problem must define its Target and Predictor variables. The Unsupervised problem must define its operational features and target insights. Both must establish clear baseline viability derived from the EDA.	
Milestone 2 Supervised DM	● Validation Rigor & Leakage Prevention: Implementation of an explicit validation framework that strictly blocks data leakage. Crucial: Students must account for spatial or temporal auto-correlation (e.g., using a time-based train/test split like pre-2019 vs. post-2019 rather than random k-fold cross-validation, which leaks temporal macrotrends). ● Class Imbalance Mitigation: Concrete algorithmic strategies to address the database's heavy-tailed distributions (e.g., millions of minor Class A/B fires vs. few catastrophic Class G fires, or sparse categories in NWCG_GENERAL_CAUSE). Proper use of SMOTE, downsampling, class weighting, or stratified evaluation metrics. ● Model Benchmarking & Metrics: Evaluation of at least three distinct algorithmic families (e.g., baseline regularized linear/logistic regression, random forests, and gradient boosted trees like XGBoost/LightGBM). Evaluation must transcend basic Accuracy and utilize highly informative, matrix-aware metrics: ROC-AUC, Precision-Recall curves, F1-scores, and localized Confusion Matrices.	25
Milestone 3 Unsupervised DM	● Feature Preprocessing & Space Setup: Rigorous mathematical handling of features before feeding them to unsupervised algorithms. This includes standardizing continuous metrics, handling high-cardinality categorical data, and encoding cyclical features (e.g., converting Day-of-Year or DISCOVERY_TIME to sine/cosine coordinates to preserve distance boundaries). ● Algorithmic Mastery: Flawless execution of a clustering pipeline (e.g., K-Means, DBSCAN, HDBSCAN, or Gaussian Mixture Models) or an Association Rule pipeline (e.g., Apriori or FP-Growth). If clustering is chosen, teams must use a validation metric (e.g., Silhouette Analysis, Davies-Bouldin index, or Elbow plots) to justify the hyperparameter selections (k, ε, etc.). If Association Analysis is chosen, support, confidence, and lift thresholds must be mathematically defended.	25

	• Insight Synthesis & Evaluation: Translation of mathematical clusters or discovered rules into operational realities. Teams must demonstrate how their unsupervised outputs map to historical anomalies, geographic "fire complexes," or administrative reporting patterns without natively utilizing text labels.	
Engineering & Optimization	• Feature Engineering Depth: Construction of sophisticated engineered columns built from the spatial-temporal primitives (e.g., "Days elapsed between discovery and containment", "Ignition distance from regional bounds", or grouping raw causes into customized risk classes). • Hyperparameter Search & Tuning: Execution of a systematic optimization pass (e.g., Grid Search, Random Search, or Bayesian Optimization via Optuna) between Milestones 2/3 and the Results Freeze date. • Code Architecture & Reproducibility: Delivery of a clean, structured repository. Clean notebooks, modular function definitions, explicit handling of data paths, a fully operational requirements.txt or environment.yml configuration file, and no absolute hardcoded directories.	15
Final Presentation & Demo	• Live Code Execution & Pipeline Demo: A fully operational, end-to-end live execution of the code engine during class. The demo must clearly load a test subset, perform data transformations, fetch predictions from the saved model weights, or visualize the cluster assignments in real-time. Code blocks must be cleanly commented and run error-free. • Technical Synthesis: A concise, 10-to-15 minute narrative detailing the data story. Slides are optional, but the verbal defense must rigorously explain why specific engineering and algorithmic tradeoffs were chosen, mapping directly back to the initial patterns identified in Milestone 1.	15
Total Points		100

Universal Quality Constraints & Point Deduction Rules

• Late Submission Policy: Submissions encounter an automatic deduction penalty of -10% per 24-hour cycle past the posted deadline. Assignments are capped at a maximum of 3 late days (72 hours), after which a flat score of zero is logged.

• Plagiarism & Hardcoded Fabrications: Codebases utilizing completely hardcoded score parameters or presenting copied material from online repositories receive an immediate score of zero points and are directly escalated to the SCU Academic Integrity Board for administrative investigation.